
DAPHNE Dead-Time Models and

Grouped Hermes

Model definitions, full-chain RTL simulation, and current resource status

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DEEP UNDERGROUND
NEUTRINO EXPERIMENT



Argument

Observed: at 4.6 kHz/channel, the present self-trigger path loses about 7.6 % of generated triggers, or about 350 trigger/(s channel).

Mitigation path

1. Shorten the record: reduce per-channel builder occupancy.
2. Add ring buffering: decouple trigger acceptance from immediate waveform packing.
3. Decide the output contract: overlap is easy to accept; non-overlap requires coalescing.

Caution: coalescing is promising as a non-overlap policy, but the current output-full gate makes it a backpressure problem, not yet an HDL performance result.

Observable And Model Boundary

Observable

$$D(\lambda) = 1 - \frac{N_{\text{accepted}}(\lambda)}{N_{\text{generated}}(\lambda)}$$

Accepted means the trigger passed the channel-local rule. It may still wait in the readout path.

Simulation layers

- ▶ C++ model: fast stochastic ranking of frame policies.
- ▶ RTL wrapper: live builders and muxes under synthetic triggers.
- ▶ Hermes chain: real `tx_mux` ready/backpressure path.
- ▶ FPGA build: resource, routing, and timing feasibility.

Model Definitions I: Fixed Records And Rings

Model	Meaning in this deck
Legacy 1024	Old fixed waveform model. One trigger emits a 1024-sample record, modeled as a long per-channel busy interval and 232 output words.
512 waveform	Same fixed-record contract, but a 512-sample record and 120 output words. This is the first-order dead-time reduction.
Ring0	Per-channel 2048-sample ring plus descriptor queue. Trigger acceptance is decoupled from immediate waveform packing, but same-channel spacing is still one full frame.
Ring50	Ring model with relaxed same-channel spacing. For 512-sample records, <code>signal_delay_steps=16</code> allows about 256 samples of overlap.
1024 ring max	Same idea for 1024 in the C++ model. The existing 5-bit delay control maxes at 31, so the largest tested overlap is 496 samples, not exactly 512.

Important: accepting overlapping triggers and emitting non-overlapping waveforms are different requirements.

Model Definitions II: Coalesced And Tail

Model	Meaning in this deck
Coalesced, current gate	C++ variable-interval model. Nearby trigger windows merge into one emitted physical interval, but the old output-full gate is still applied.
Coalesced, relaxed gate	Same C++ interval model with the output gate opened. This is an upper-bound study, not current RTL.
<code>coal-tail512</code>	RTL fixed-fragment continuation model. It preserves non-overlap at Ethernet by chaining 512-sample fragments, but it is not true variable-length coalescing.

Boundary

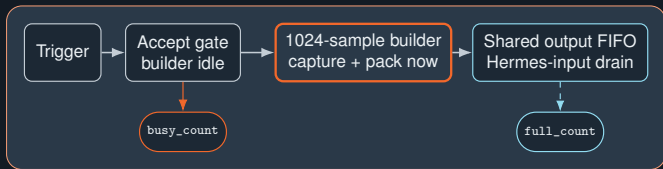
Coalescing changes the unit of output. It is not only a trigger-acceptance rule; it also changes packet assembly and metadata ownership.

Model Definitions III: Grouped Hermes

Model	Meaning in this deck
Grouped+Hermes 5x8	Full-chain RTL wrapper: 40 live <code>stc3_frame_source</code> channels, 5 serializers, 8 channels per serializer, real Hermes <code>tx_mux</code> , <code>READY_AWARE</code> enabled, 2048-word Hermes input buffers.
Grouped+Hermes 10x4	Same full-chain RTL wrapper, but 10 Hermes sources with 4 channels each. It reduces serializer contention in simulation but costs more BRAM/routing in FPGA implementation.

Current RTL note: the live grouped-Hermes branch is 512-sample based. The 1024 comparison is C++ only unless the RTL constants are parameterized and rebuilt for 1024.

Current Workflow



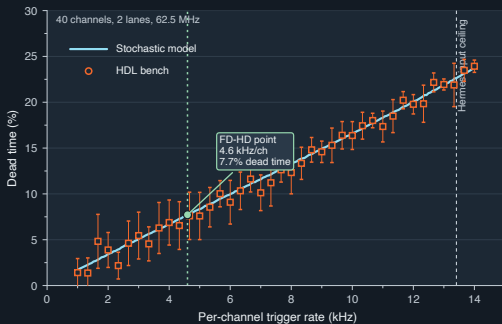
Where loss starts

$$B = 1037 \text{ clocks}$$

$$\tau_b = 16.592 \mu\text{s}$$

A second same-channel trigger during this interval increments `busy_count`.

Baseline Data



The FD-HD point is already on the rising part of the firmware dead-time curve.

Study Workflow

1. Establish the real firmware baseline from successful builds and routed reports on K26.
2. Use the C++ dead-time model to rank architecture ideas over the full rate range quickly.
3. Wrap the live RTL in the mezzanine dead-time bench to test the current branch, not a behavioral copy.
4. Use synth/impl only on the narrowed HDL candidates that survive the C++ and RTL filters.

Why this matters

The expensive step is implementation on the Kria PL. The workflow is designed to spend iteration time first on model ranking, then on RTL behavior, and only then on place-and-route.

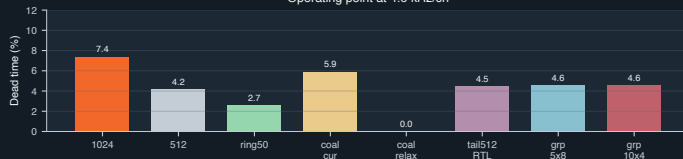
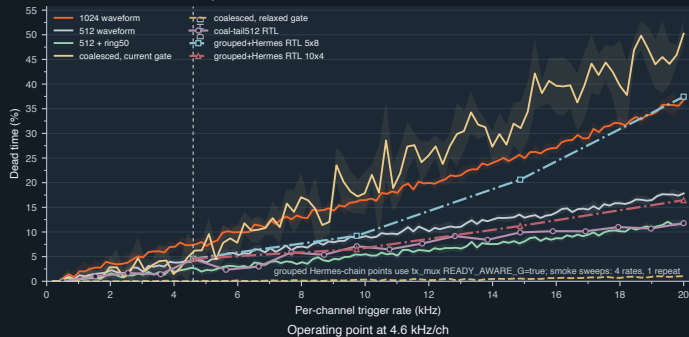
Why This Became A Rabbit Hole

1. **Shorter fixed records:** $1024 \rightarrow 512$. This helps first and keeps the old output contract.
2. **Ring with relaxed spacing:** `ring50` accepts more triggers, but it does so by allowing overlapping waveform records.
3. **Non-overlap redesign:** once overlap is forbidden, the problem moves from trigger acceptance into the frame builder and output seam.

Why this became confusing

Dead-time mitigation and waveform-contract preservation are not the same problem. The moment overlap is forbidden, trigger acceptance is no longer the whole story: the frame builder and output seam become the real bottleneck.

Grouped Hermes-chain RTL versus earlier dead-time architectures



Nominal Hermes load near 20.0 kHz/ch

arch	dead	Hermes	arch	dead	Hermes
1024	36.8%	76.0%	coal relax	1.0%	57.1%
512	17.8%	51.3%	tail512 RTL	11.8%	52.8%
512+ring50	11.8%	55.1%	grp 5x8	37.4%	37.3%
coal cur	50.2%	28.5%	grp 10x4	16.5%	50.1%

Hermes column is word-budget use; grouped values are measured at tx_mux output, others are estimated from sent block words. grouped RTL ready-low cycles: 0, 0

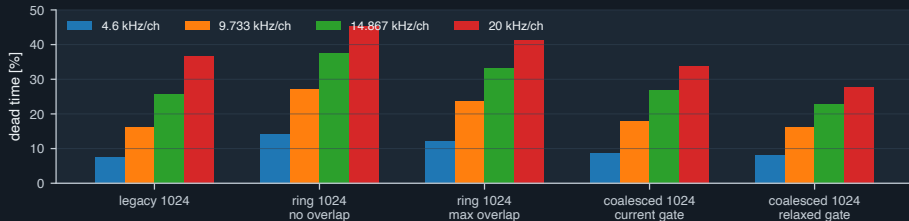
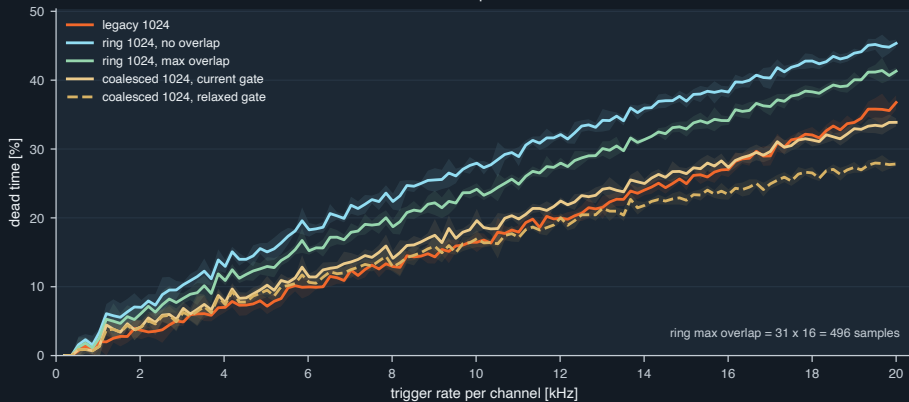
Latest Full-Chain RTL Points

Architecture	Sources	Ch/source	4.6 kHz	9.733 kHz	20 kHz
Grouped+Hermes 5x8	5	8	4.6 %	9.2 %	37.4 %
Grouped+Hermes 10x4	10	4	4.6 %	6.5 %	16.5 %

Architecture	Hermes load at 4.6 kHz	9.733 kHz	14.867 kHz	20 kHz
Grouped+Hermes 5x8	13.7 %	27.6 %	36.5 %	37.3 %
Grouped+Hermes 10x4	13.7 %	28.5 %	40.9 %	50.1 %

Both RTL sweeps used the real Hermes `tx_mux` ready path with no artificial Ethernet or UDP stalls. Source ready-low cycles were zero in these nominal runs.

1024-sample variants



Methodology Stack

C++ model

- ▶ fast dense sweeps
- ▶ architecture ranking
- ▶ reject-cause accounting
- ▶ transport sensitivity studies

RTL wrapper

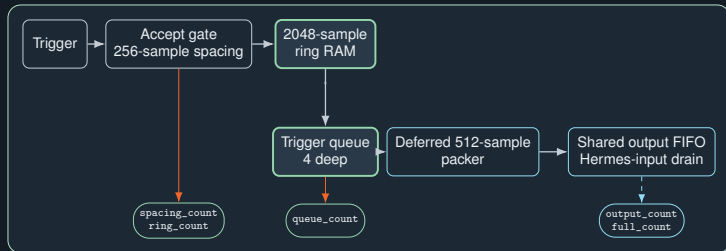
- ▶ live builder + mux
- ▶ exact branch interface
- ▶ full-spectrum dead-time sweep
- ▶ catches contract mistakes

Implementation

- ▶ K26 resource fit
- ▶ timing closure
- ▶ packaging path
- ▶ real feasibility filter

Discipline: C++ ranks ideas, RTL validates branch behavior, implementation decides whether the idea survives on hardware.

Why Ring50 Helps

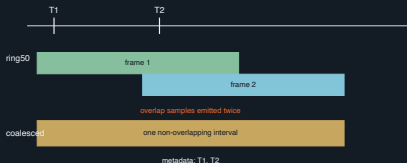


Mechanism: trigger metadata waits while samples remain in RAM; the half-frame spacing rule accepts closer triggers.

Consequence: output records can overlap in waveform time.

Why Coalesced Can Work

Same two triggers, two output policies



- ▶ `ring50` accepts more by emitting several fixed records, even if they cover the same physical waveform.
- ▶ `coalesced` accepts more without overlap because nearby triggers share one emitted waveform interval.
- ▶ That reduces duplicate waveform words, but only if the assembler and output contract also change.

What We Tested: `coal-tail512`

Policy

- ▶ 2048-sample ring per channel
- ▶ queue depth 4
- ▶ fixed 512-sample records
- ▶ no Ethernet-visible overlap
- ▶ chained continuation when the frame end is still active

Boundary: this is still fixed-record transport, not a variable-length coalescer.

HDL seam cleanup

- ▶ acceptance no longer uses inherited `prog_empty/prog_full`
- ▶ builder/readout seam is explicit
- ▶ packet availability is tracked as count
- ▶ continuation and trigger offset are explicit

Why `coal-tail512` Is Different From Coalesced

`coal-tail512`

- ▶ fixed 512-sample fragments
- ▶ no overlap visible on Ethernet
- ▶ if the waveform is still active, open another fixed fragment

Coalesced

- ▶ one variable-length waveform interval
- ▶ no overlap visible on Ethernet
- ▶ nearby triggers extend or merge the same interval

Practical difference

`coal-tail512` keeps the fixed record size and fixes the seam. Coalesced changes the unit of output from “one fixed fragment” to “one physical interval”.

Build And Routing Status

Metric	5-source legal route	10-source illegal route
CLB LUTs	99 560 (85.0 %)	109 271 (93.3 %)
Registers	165 056 (70.5 %)	177 485 (75.8 %)
BRAM tiles	103/144 (71.5 %)	128/144 (88.9 %)
Bitstream	yes	no, route illegal

- ▶ 5-source branch: grouped-hermes-resource-build-candidate, commit 77682da; full K26C bitstream produced and timing met with WNS 0.053 ns.
- ▶ 10-source/2048-buffer variant: dead-time simulation is better, but bitgen failed with 142 601 unrouted signals and 234 221 node overlaps.

Current RTL Result: Grouped Hermes

- ▶ The 5-source grouped-Hermes design is the current buildable resource candidate with legacy 2048-word Hermes input buffers.
- ▶ The 10-source version reduces dead time because each serializer serves fewer channels, but it is routing-limited on the full K26C image.
- ▶ In the nominal full-chain RTL sweeps, Hermes `source_ready` did not assert backpressure. The measured dead time is therefore dominated by source/serializer admission, not by Hermes drain.

Practical candidate

Use 5-source/2048-buffer grouped Hermes for buildable firmware work. Use 10-source results only to guide resource-reduced or floorplanned variants.

What The Latest Sweeps Are Saying

- ▶ The old 1024-sample family is not the right direction. Even the best 1024 C++ case is 27.8% dead time at 20 kHz/channel.
- ▶ The useful dead-time gains come from 512-sample records, shorter source occupancy, and reducing serializer fan-in.
- ▶ Full-chain Hermes simulation changes the story: the grouped 5x8 path is buildable but has 37.4% dead time at 20 kHz/channel; 10x4 drops that to 16.5% but does not currently route.
- ▶ No nominal Hermes ready backpressure was observed, so Hermes ready logic is not the main performance lever at this operating point.

Therefore

The current buildable target is 5-source grouped Hermes with 2048 input buffers. The next target is lowering source/serializer cost enough that a lower-fan-in design can still route.

Why Coalesced Stays On The Roadmap

- ▶ Coalesced is still the clean non-overlap end-state: one physical waveform interval should be emitted once, not as several fixed fragments.
- ▶ The C++ model says it can be excellent, but only when the output gate and variable-length transport are redesigned together.
- ▶ The grouped-Hermes work is orthogonal: it changes source fan-in and the Hermes input topology, while coalescing changes the waveform-output contract.
- ▶ The practical order is to keep the 5-source buildable baseline stable, then test resource-reduced ways to lower serializer fan-in, then revisit true interval transport.

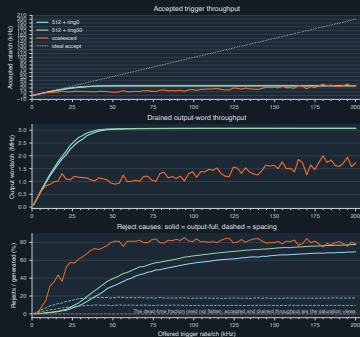
What the relaxed-gate curve means

0.0% is an acceptance-side upper bound from the C++ study. It motivates the roadmap; it is not today's RTL result.

Next HDL-Relevant Questions

1. Can the 5-source buildable point be made faster by simplifying self-trigger and CFD logic without changing the packet contract?
2. Can a lower-fan-in topology be made routable by shrinking Hermes input buffers, floorplanning, or reducing replicated source logic?
3. What is the smallest useful source count between 5 and 10 once resource and routing cost are included?
4. If we later move to true coalescing, what metadata policy preserves multiple peak descriptors in one non-overlap waveform interval?

Backup: Why The Wide Curve Does Not Flatten



Backup: The Fraction Keeps Rising By Definition

$$D(\lambda) = 1 - \frac{A(\lambda)}{G(\lambda)}$$

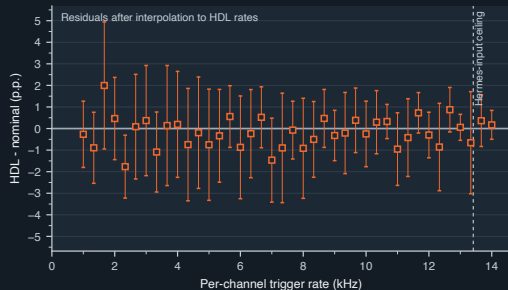
Interpretation: When accepted throughput $A(\lambda)$ saturates but generated rate $G(\lambda)$ keeps increasing, the dead fraction continues to rise. The quantities that flatten are accepted record rate and drained output-word rate.

Transport scale:

$$\frac{62.5 \text{ MHz}}{20} \approx 3.125 \text{ Mword}/(\text{s channel})$$

For 120 word records, this corresponds to about 25.6 kHz/channel.

Backup: Model Validation



Mean absolute residual	0.83 p.p.
Maximum absolute residual	2.97 p.p.
Mean HDL spread	1.20 p.p.
Maximum HDL spread	3.81 p.p.

The stochastic model is adequate for ranking options. New frame policies still need HDL spot checks.

Backup: Acceptance Rules I

Legacy

$$\text{accept}(i, t) \iff t \geq b_i \wedge \neg \text{progFull}_i(t), \quad b_i \leftarrow t + \tau_b.$$

Ring

$$\text{accept}(i, t) \iff \text{spacingOK} \wedge \text{queueOK} \wedge \text{ringOK} \wedge \neg \text{progFull}_i(t).$$

For ring50, spacingOK is a 256 sample same-channel window for a 512 sample record.

Backup: Acceptance Rules II

Coalesced probe

$$\text{accept}(i, t) \iff \text{queueOK} \wedge \text{ringOK} \wedge \neg \text{progFull}_i(t).$$

Overlapping trigger windows extend or join one queued output interval.

Backup: Data Provenance

- ▶ Simulation repo: `daphne_mezz_xc_sim`, branch `marroyav/deadtime-tail512-study`, commit `36073b1`.
- ▶ Firmware branch for grouped-Hermes RTL:
`marroyav/grouped-hermes-resource-build-candidate`, build commit `77682da`.
- ▶ Full-chain RTL CSVs: `deadtime_grouped_hermes_src5_ch8_hdl.csv` and `deadtime_grouped_hermes_src10_ch4_hdl.csv`.
- ▶ 1024 C++ CSVs: `deadtime_1024_..._cpp.csv`; these are not full-chain RTL because the live source/serializer constants are currently 512.
- ▶ Latest plots copied from `data/output/plots` on May 14, 2026:
`deadtime_grouped_current_arch_compare` and `deadtime_1024_variant_compare`.